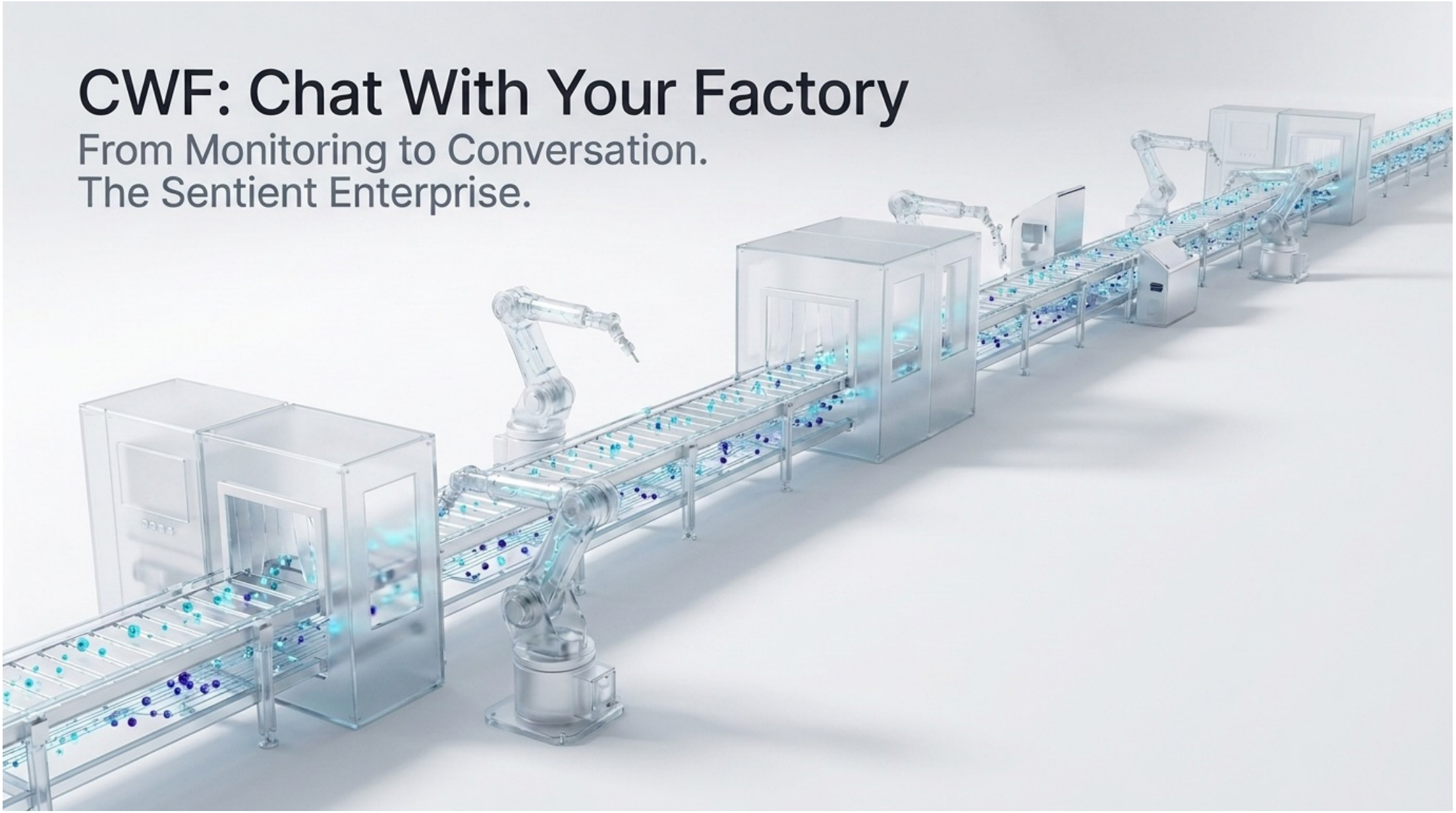
# CWF: Chat With Your Factory

## From Monitoring to Conversation.
## The Sentient Enterprise.



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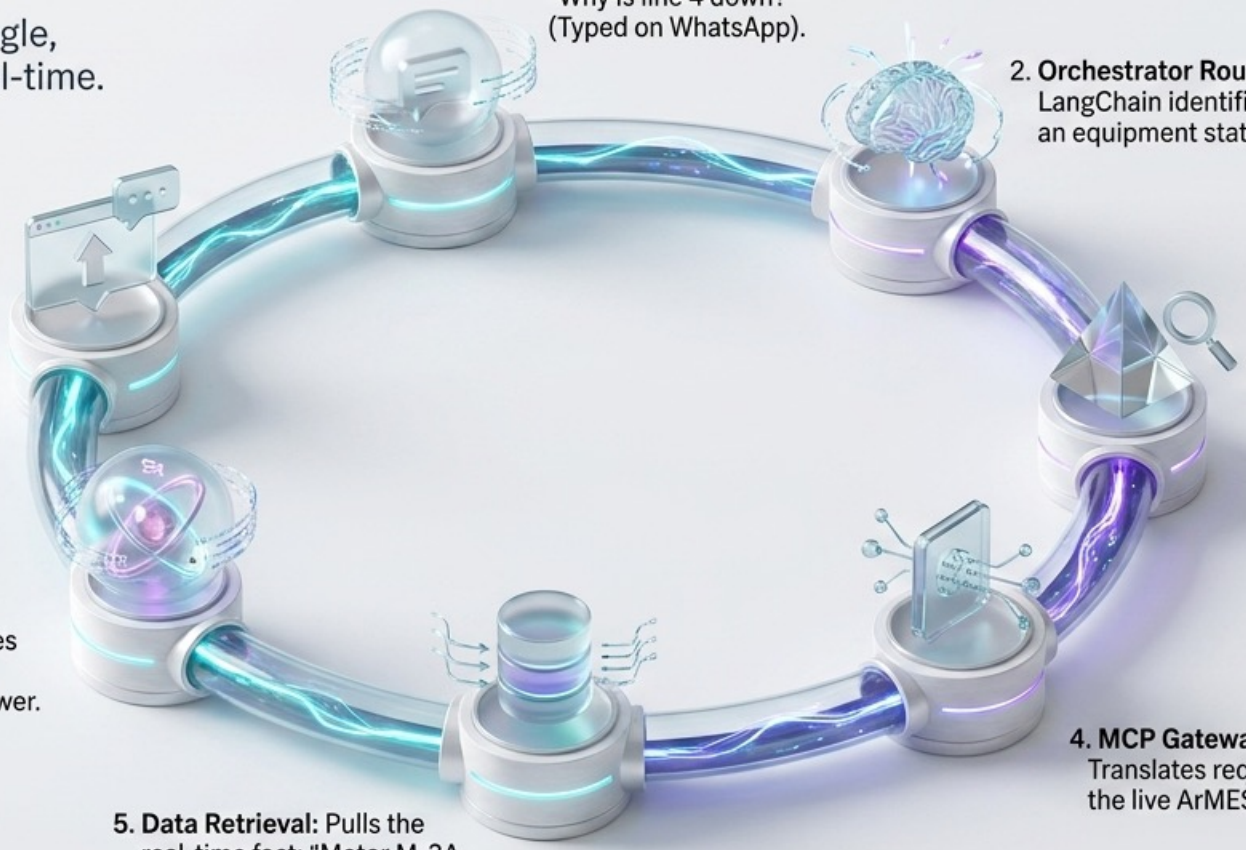
## The Modality Matrix

| Mobile App | Desktop Web | WhatsApp |
|---|---|---|
| **Environment:** On the floor | **Environment:** In the office | **Environment:** Remote/Transit |
| **Primary Input:** Voice | **Primary Input:** Deep Dive/Dashboard | **Primary Input:** Quick Chat |
| **Cognitive Load:** Minimal (Hands-free) | **Cognitive Load:** High (Analytical) | **Cognitive Load:** Low (Instant update) |

# The Cognitive Loop: From Query to Insight

Tracing the lifecycle of a single, real-world interaction in real-time.



1. **Natural Language Query:** "Why is line 4 down?" (Typed on WhatsApp).
2. **Orchestrator Routing:** LangChain identifies this as an equipment status intent.
3. **Semantic Search:** Vector DB retrieves manuals regarding Line 4's specific machinery.
4. **MCP Gateway API Call:** Translates request to ping the live ArMES database.
5. **Data Retrieval:** Pulls the real-time fact: "Motor M-3A over-temp alert active."
6. **LLM Synthesis:** Combines the live data and manual context into a human answer.
7. **Multi-Modal Response:** "Line 4 is down due to Motor M-3A overheating. Would you like me to generate a maintenance work order?"